$$n = \frac{m}{M}, \qquad C = \frac{n}{V}$$

$$C_1 V_1 = C_2 V_2$$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

$$pH = -\log_{10} a_{H^+}$$

$$pK_a = -\log_{10} K_a$$

$$K_w = [H^+][OH^-]$$

$$pH + pOH = pK_w$$

$$pH = pK_a + \log_{10} \frac{[A^-]}{[HA]}$$

$$\frac{[A^-]}{[HA]} = 10^{pH - pK_a}$$

$$pH = 6.1 + \log_{10} \frac{[HCO_3^-]}{0.03 P_{CO_2}}$$

$$f_{A^-} = \frac{1}{1 + 10^{pK_a - pH}}$$

$$f_{BH^+} = \frac{1}{1 + 10^{pH - pK_a}}$$

$$Z = \sum_j q_j f_j$$

$$Z(pI) = 0$$

$$pI = \frac{pK_{a,i} + pK_{a,i+1}}{2}$$

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G = \Delta G^{\circ\prime} + RT \ln Q$$

$$\Delta G^{\circ\prime} = -RT \ln K'_{eq}$$

$$\Delta G_\Sigma = \sum_i \Delta G_i$$

$$\Delta G = -nF\Delta E$$

$$\Delta G_{1\to 2} = RT \ln \frac{C_2}{C_1} + zF(\psi_2 - \psi_1)$$

$$\Delta G_{1\to 2} = 0 \;\Rightarrow\; \psi_2 - \psi_1 = \frac{RT}{zF} \ln \frac{C_1}{C_2}$$

$$\pi = iCRT$$

$$Y = \frac{pO_2}{P_{50} + pO_2}$$

$$Y = \frac{(pO_2)^{n_H}}{P_{50}^{n_H} + (pO_2)^{n_H}}$$

$$\log_{10} \frac{Y}{1-Y} = n_H \log_{10} \frac{pO_2}{P_{50}}$$

$$K_d = \frac{[Ab][Ag]}{[Ab:Ag]}$$

$$\theta = \frac{[L]}{K_d + [L]}$$

$$v_0 = \frac{V_{max}[S]}{K_m + [S]}$$

$$V_{max} = k_{cat}[E]_T$$

$$K_m = \frac{k_{-1} + k_{cat}}{k_1}$$

$$K_d = \frac{k_{-1}}{k_1}$$

$$v_0 \underset{[S] \ll K_m}{\approx} \frac{k_{cat}}{K_m}[E]_T[S]$$

$$[S] = K_m \;\Rightarrow\; v_0 = \frac{V_{max}}{2}$$

$$[S] \gg K_m \;\Rightarrow\; v_0 \approx V_{max}$$

$$\frac{1}{v_0} = \frac{K_m}{V_{max}}\frac{1}{[S]} + \frac{1}{V_{max}}$$

$$\alpha = 1 + \frac{[I]}{K_i}, \qquad \alpha' = 1 + \frac{[I]}{K_i'}$$

$$v = \frac{V_{max}[S]}{\alpha K_m + \alpha'[S]}$$

$$V_{max,app} = \frac{V_{max}}{\alpha'}$$

$$K_{m,app} = \frac{\alpha}{\alpha'}K_m$$

$$K_i = \frac{[I]}{\alpha - 1}$$

$$n \in \{4, 6, 8, \ldots\}$$

$$N_\beta = \frac{n}{2} - 1$$

$$N_{acetyl} = \frac{n}{2}$$

$$N_{NADH,\beta} = N_{FADH_2,\beta} = \frac{n}{2} - 1$$

$$ATP_{net} \approx 10\frac{n}{2} + 4\left(\frac{n}{2} - 1\right) - 2$$

$$ATP_{net} \approx 7n - 6$$

$$ATP_{acetyl} \approx 3(2.5) + 1.5 + 1 = 10$$

$$ATP_{glucose} \approx 4 + 10(2.5) + 2(1.5) = 32$$

$$ATP_{glucose} \approx 4 + 8(2.5) + 4(1.5) = 30$$

$$R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$$

$$F = 96485\,\mathrm{C\,mol^{-1}}$$

$$N_A = 6.02214076 \times 10^{23}\,\mathrm{mol^{-1}}$$

$$T_K = T_{\circ C} + 273.15$$